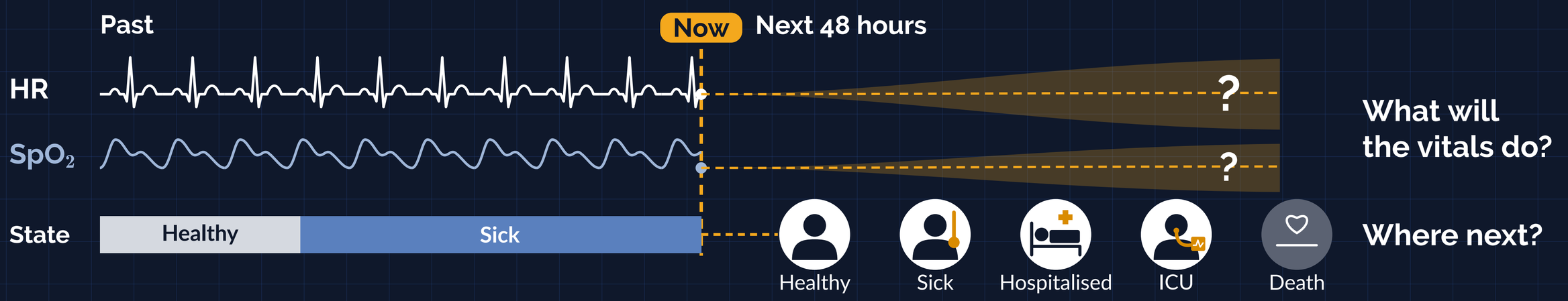
nMSM: A Neural Multi-State Model for Joint State–Covariate Trajectory Forecasting

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Code & paper

Which state will you be in next?



1 Most work predicts a single event

Survival analysis
the time until one event.

Alive → when? → Event

Real patients move between states, worsen and recover, while vitals change.

Healthy ⇄ Sick ⇄ Hospital → Death

2 Our goal

One model that predicts the next **state** and the next **vitals** together, so we can **forecast** a patient and ask **what if** we treat.

3 How nMSM works

H_t: history up to hour t S: state V: vitals

$$p(S_{t+1}, V_{t+1} \mid H_t) = \underbrace{p(S_{t+1} \mid H_t)}_{\text{next state}} \; \underbrace{p(V_{t+1} \mid H_t, S_{t+1})}_{\text{next vitals, given that state}}$$

History H_t — past states and vitals

→ **Shared encoder** — Transformer or GRU summarises the history

→ **State head** — next state, only allowed moves

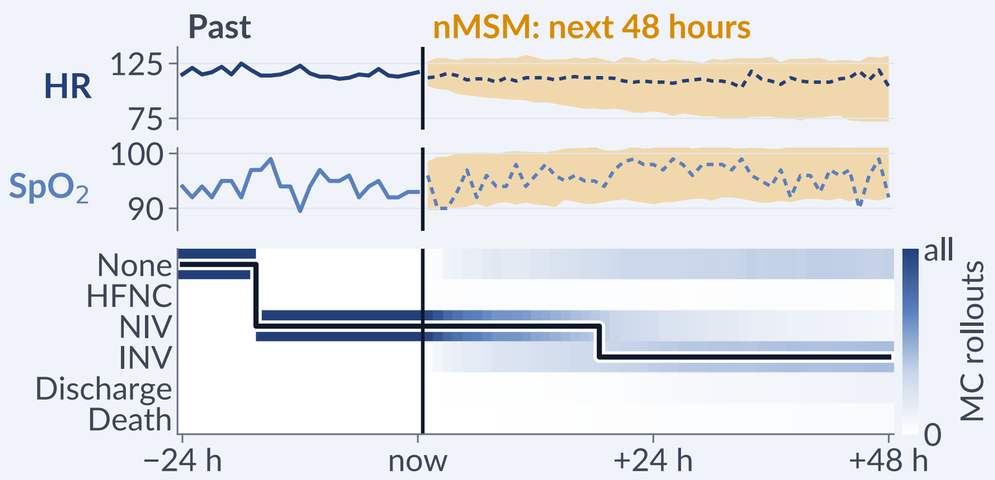
→ **Vitals head** — next vitals, given that state

→ **New hour** S_{t+1}, V_{t+1}

Append and repeat: an MC rollout gives many possible futures

4 Results: forecast and what if

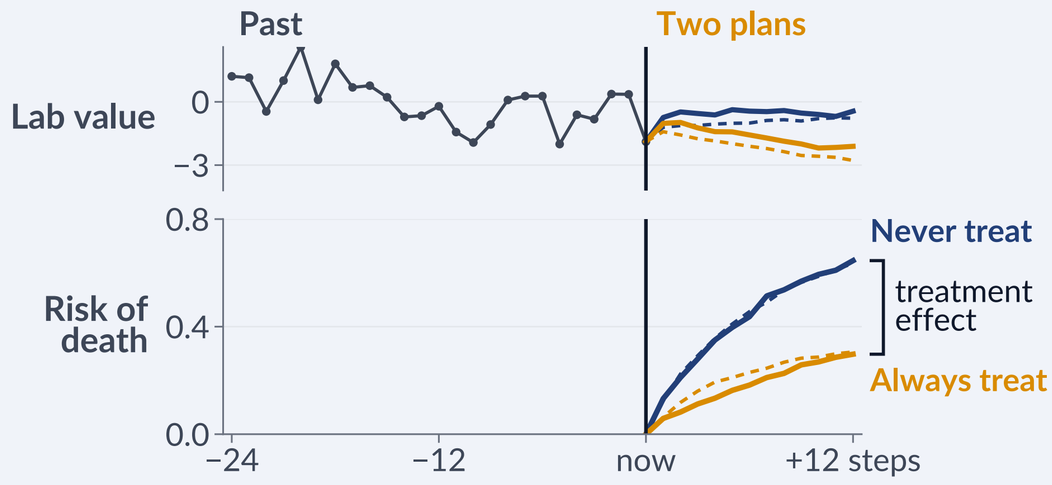
Trajectory forecasting — next states and vitals



— recorded ···· what happened nMSM range — recorded state

One ICU stay; shading = share of MC rollouts. States are breathing support: **HFNC** high-flow oxygen, **NIV** mask ventilation, **INV** breathing tube + ventilator.

Counterfactual estimation — what if we treat?



— nMSM ···· true answer (simulator)

One simulated patient with a simulated lab value, two treatment plans fixed in advance. nMSM follows the true answer.

5 Data and accuracy

Three cohorts

ICU · MIMIC-IV
50,873 stays · hourly

Hospital · SAIL
28,321 people · quarterly

Simulated
8,000 trajectories · known truth

Forecast accuracy

	error ↓		AUC ↑	
	others	nMSM	others	nMSM
Synthetic	.1588	**.1430**	.7712	**.8195**
MIMIC-IV	.0644	**.0617**	.8668	**.8902**
SAIL	.0757	**.0685**	.6974	**.7255**

Error: IBS; AUC: iAUC. Means of 5 folds. Others: best of 5–6 other models.

What-if accuracy

	error ↓		
	states	vitals	effect
Linear	.1068	.3710	.0502
G-Transformer	.0089	**.0729**	.0287
nMSM	**.0087**	.0898	**.0033**

128 simulated patients with known answers. Effect: change in death risk, always vs never.

Machine Learning for Health (ML4H) 2026 – Data: synthetic illness–death, MIMIC-IV (PhysioNet, credentialed), SAIL Databank (secure environment) – Code: github.com/TheMaxmillanRies/nmsm-ml4h